Semester V **B. Tech CSE (AI)**

BCSAI 502: PROLOG PROGRAMMING

3L + 0T + 0P + 3C **MM 100**

Unit 1: An overview of Prolog

An overview of Prolog, components of a Prolog program, syntax of Prolog, meaning of a Prolog program, Clauses, Programs and Queries

Unit 2: Working with Lists, Numbers and Operators

Notation, Head and Tail, Some Built-in Predicates for List Manipulation, working with numbers:The is-Operator for Arithmetic Evaluation, Predefined Arithmetic Functions and Relations, workimg with operators: Precedence and Associativity, Declaring Operators with op/3.

Unit 3: Backtracking, Cuts and Negation

Backtracking and Cuts: Backtracking Revisited, Problems with Backtracking, Introducing Cuts, Problems with Cuts, Negation as Failure: The Closed World Assumption, Horn Formulas and Resolution.

Unit 4: Logic Foundations of Prolog

Translation of Prolog Clauses into Formulas ,Horn Formulas and Resolution.

Unit 5: Recursive Programming

Induction in Mathematics, The Recursion Principle ,What Problems to Solve Debugging .

Text Books:

1. PROLOG Programming for Artificial Intelligence Bratko I., Addison-Wesley, Reading, MA, 1986

Reference Books:

1. Programming in Prolog , Clocksin W.F. & Mellish C.S., Springer-Verlag, Berlin, 1981, A good introduction to programming in PROLOG
2. Foundations of Logic Programming, Lloyd J.W., Springer-Verlag, Berlin, 1984, Logic programming is a theory behind the PROLOG. You can find introduction to this topis here.